Package luahyperbolic

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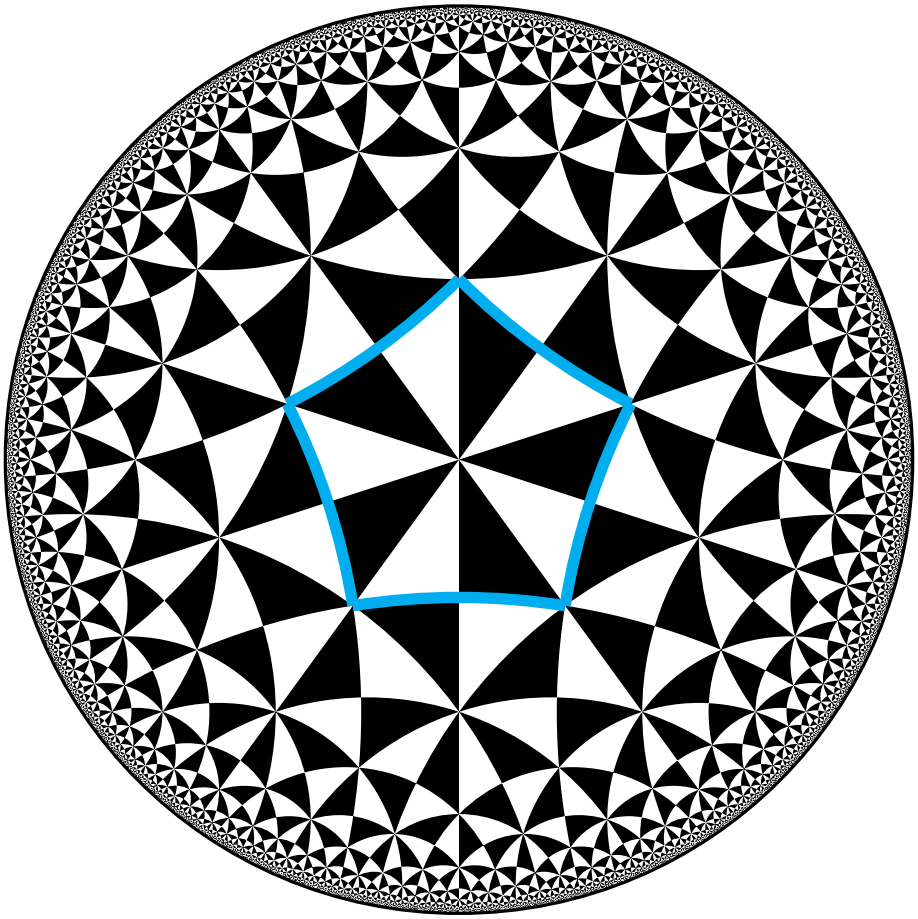

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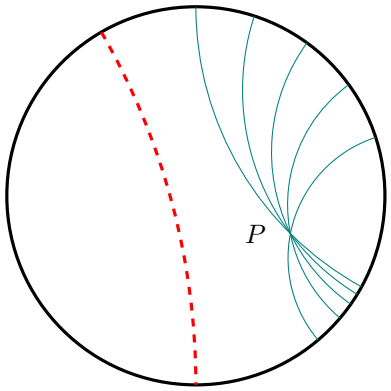
1 Introduction

1.1 What you can do with this package

1. Draw hyperbolic geometry pictures, high precision (pdf output).
2. Use an easy syntax. At least, easier than with LaTeXmacros and blody pgf.
3. Export generated pictures to external tikz files if you want.
4. Read the source code and learn hyperbolic geometry.
5. Compile **without** installing python packages or lua packages etc.
6. Compile **without** "shell-escape" mode.
7. Compile everything in a single pass. No (Leeloo Dallas) multipass. Want to illustrate Euclid's fifth postulate ? Do it right there in your tex file without having to include external files :

Euclid's 5th Postulate

```
\begin{luacode*}
hyper.tikzBegin("scale=2.5")
local P = complex(0.5,-0.2)
local A = complex.exp_i(math.pi/10)
for k=1,5 do
  hyper.drawLine(P, A^k, "teal")
end
hyper.labelPoint(P, "$P$", "left=.2cm")
local style = "very thick, dashed, red"
hyper.drawLine(complex.J,-complex.I,style)
hyper.tikzEnd("euclid5thPostulate.tikz")
\end{luacode*}
```



1.2 What this package cannot do for you

1. Really intensive computations, millions of geodesics etc. Source code is not optimized to keep maximum readability. All computations involve automorphisms. See other remarks below.
2. Coffee

1.3 Efficiency

For heavy computations, please use a ~~Python~~ C library. Nevertheless, this library is still reasonably efficient and usable:

- The first tiling illustration of this documentation (hyperbolic 2-4-5 tiling with 2361 geodesics) compiled in 2.7 seconds on a 2020 intel macbook air laptop.
- This entire documentation including all pictures compiles in approx. 15 seconds on the same machine, the majority of the time weing spent on the tilings. The package is suitable for producing lecture notes in hyperbolic geometry, which was the reason for all of this. For an entire book with dozens of pictures, you may need to externalize some of the pictures.
- Tests show that the bottleneck is TikZ, not Lua. In a future version, some future maintainer may add other fontends to the library, including drawing directly to the pdf with pdf primitives wich are several orders of magnitude faster than tikz.

1.4 How to load the package

Local install

Simply put the `luahyperbolic.sty` file in your working directory and add the following line to your tex preamble :

```
\usepackage{luahyperbolic}
```

The file `luahyperbolic.sty` **contains everything needed**, including the necessary library for complex numbers. It can be sent to colleagues by email etc.

It will declare two lua globals : `complex` and `hyper`, and define a `hyperbolicdisk` environment for convenience.

If the sty file is not in the same directory as your tex file, you can write `\usepackage{../luahyperbolic}`

CTAN

Not available yet.

1.5 How to customize the package

The `luahyperbolic.sty` is generated from three lua modules : `complex.lua`, `luahyperbolic-core.lua` and `luahyperbolic-tikz.lua`. These files are in the `lua/` subdirectory of the package repo.

To build the `luahyperbolic.sty` from lua source files, execute `lua build.lua` in the root directory of the package. This will backup the old sty file to the `backup/` directory, then buid and replace the existing sty file.

1.6 Environment hyperbolicdisk

The package provides the latex environment `\begin{hyperbolicdisk} ... \end{hyperbolicdisk}` which is defined as follows:

```
\newenvironment{hyperbolicdisk}[1][scale=3]{%
  \begin{tikzpicture}[#1]
    \draw[very thick] (0,0) circle (1);
    \clip (0,0) circle (1);
}{%
  \end{tikzpicture}
}
```

In this documentation, we prefer to use the lua functions `hyper.tikzBegin(options)` and `hyper.tikzEnd(export_file)` that enable export functionalities, see below.

1.6.1 Exporting the pictures

Simply add a filename to the closing `hyper.tikzEnd()` instruction, as below:

```
hyper.tikzEnd("myfile.tikz")
```

This will save the current picture (including `\begin{tikzpicture}[options]` and `\end{tikzpicture}`

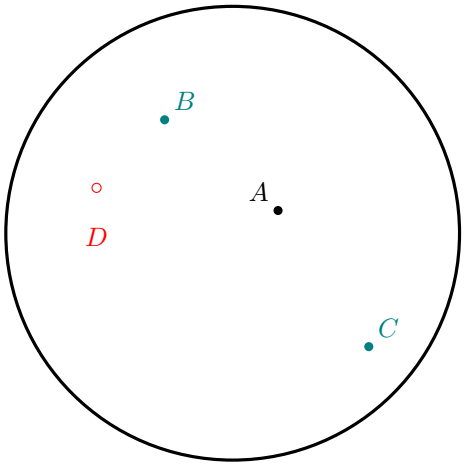
It is also possible to export the current picture at any point with `hyper.tikzExport(optional-filename)`. If no filename is provided, the package will name it `hyper_picture_N` with `N` the picture counter. Luahyperbolic keeps count of how many pictures have been produced in the document. This allows exporting images in loops, or drawing several steps of the same picture.

2 Drawing elements

2.1 Points

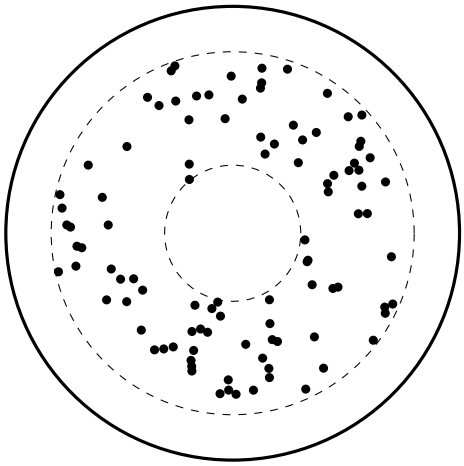
Draw and label points

```
\begin{luacode*}
local A = complex(0.2,0.1)
local B = complex(-0.3,0.5)
local C = complex(0.6,-0.5)
local D = complex(-0.6,0.2)
hyper.tikzBegin()
hyper.drawPoint(A)
hyper.drawPoints(B, C, "teal")
hyper.drawPoint(D, "white, draw=red")
hyper.labelPoint(A,"$A$") -- default: above
    ↪ left
hyper.labelPoints(B,"$B$", C, "$C$", "above
    ↪ right, teal")
hyper.labelPoint(D,"$D$","below=.4cm, red")
hyper.tikzEnd()
\end{luacode*}
```



Random points with uniform density

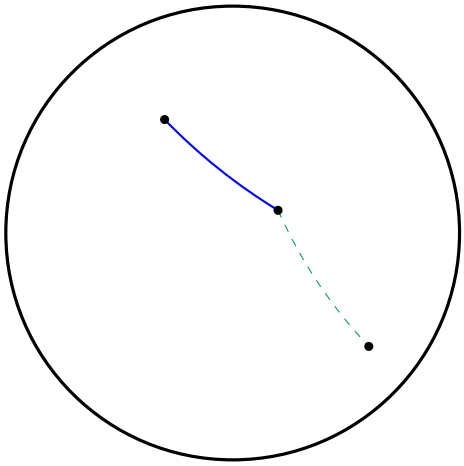
```
\begin{luacode*}
local rMin, rMax = .3, .8
hyper.tikzBegin()
for k=1,100 do
  local M = hyper.randomPoint(rMin,rMax)
  hyper.drawPoint(M)
end
hyper.tikzPrintf("\\draw[dashed] (0,0) circle
    ↪ (%s) ;", rMin)
hyper.tikzPrintf("\\draw[dashed] (0,0) circle
    ↪ (%s) ;", rMax)
hyper.tikzEnd()
\end{luacode*}
```



2.2 Geodesic segments, lines and rays

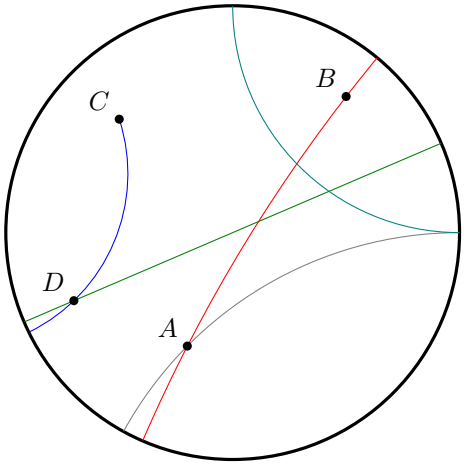
Geodesic segments

```
\begin{luacode*}
hyper.tikzBegin()
local A = complex(0.2,0.1)
local B = complex(-0.3,0.5)
local C = complex(0.6,-0.5)
hyper.drawSegment(A, B, "thick,blue")
hyper.drawSegment(A, C, "dashed,ForestGreen")
hyper.drawPoints(A,B,C)
hyper.tikzEnd()
\end{luacode*}
```



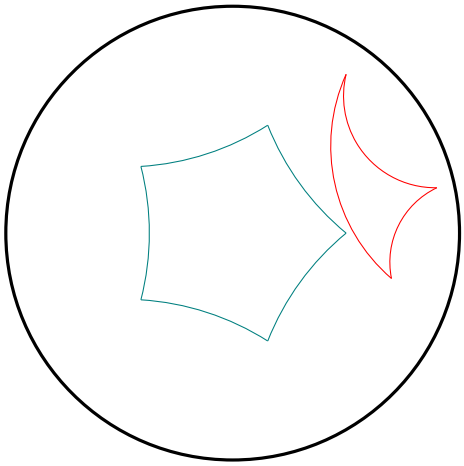
Lines and rays (through two points)

```
\begin{luacode*}
hyper.tikzBegin()
local A = complex(-0.2,-0.5)
local B = complex(0.5,0.6)
local C = complex(-0.5,0.5)
local D = complex(-0.7,-0.3)
hyper.drawLine(A, B, "red")
hyper.drawRay(C,D, "blue")
hyper.drawLine(D,-D, "green!50!black")
hyper.drawLine(A,complex(1,0), "gray")
hyper.drawLine(complex(1,0), complex(0,1), "
    ↪ teal")
hyper.drawPoints(A, B, C, D)
hyper.labelPoints(A, "$A$", B, "$B$", C, "$C$",
    ↪  D, "$D$")
hyper.tikzEnd()
\end{luacode*}
```



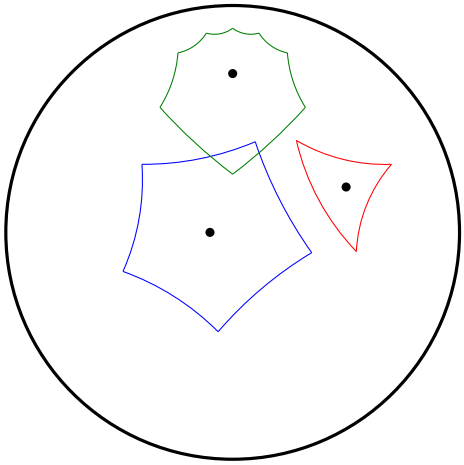
Triangles and polygons

```
\begin{luacode*}
hyper.tikzBegin()
hyper.drawTriangle(
  complex(.9,.2),
  complex(.5,.7),
  complex(.7,-.2), "red")
local points={}
for k=1,5 do
  table.insert(points, complex.polar(.5,k*2*
      ↪ math.pi/5))
end
hyper.drawPolygonFromTable(points,"teal")
hyper.tikzEnd()
\end{luacode*}
```



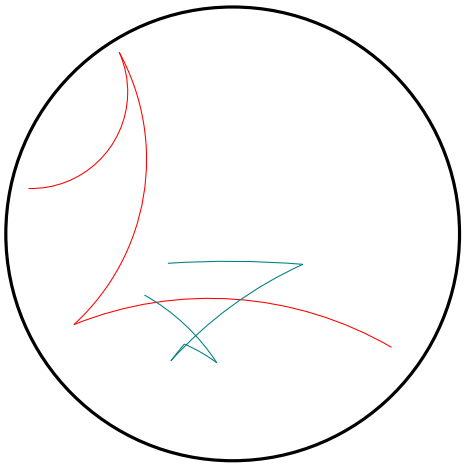
Regular polygons

```
\begin{luacode*}
hyper.tikzBegin()
local O1 = complex(.5,.2)
local A1 = complex(.7,.3)
local O2 = complex(-0.1,0)
local A2 = complex(-.4,.3)
local O3 = complex(0,.7)
local A3 = complex(0,.9)
hyper.drawRegularPolygon(O1, A1, 3, "red")
hyper.drawRegularPolygon(O2, A2, 5, "blue")
hyper.drawRegularPolygon(O3, A3, 8, "green!50!
    ↪ black")
hyper.drawPoints(O1,O2,O3)
hyper.tikzEnd()
\end{luacode*}
```



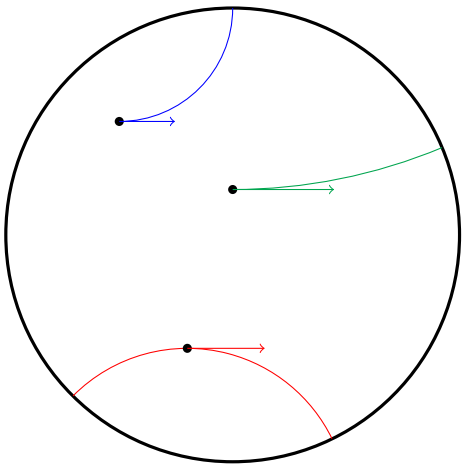
Polygonal lines

```
\begin{luacode*}
hyper.tikzBegin()
local A = complex(-.9,.2)
local B = complex(-.5,.8)
local C = complex(-.7,-.4)
local D = complex(.7,-.5)
hyper.drawPolyline(A, B, C, D, "red")
local points={}
for k=1,6 do
  table.insert(points, hyper.randomPoint(0,.7))
end
hyper.drawPolylineFromTable(points,"teal")
hyper.tikzEnd()
\end{luacode*}
```



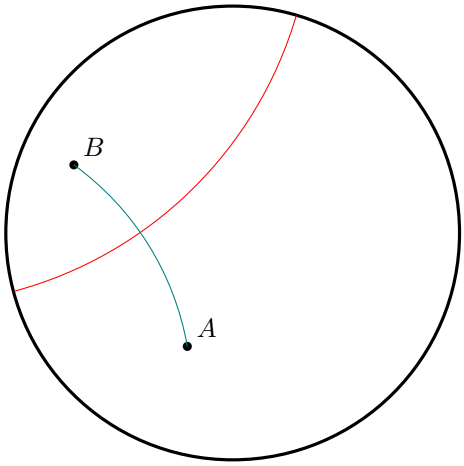
Lines and rays (point and tangent vector)

```
\begin{luacode*}
hyper.tikzBegin()
local A = complex(-0.2,-0.5)
local B = complex(-0.5,0.5)
local C = complex(0,0.2)
local v =  complex(1,0)
hyper.drawLineFromVector(A, v, "red")
hyper.drawRayFromVector(B, v, "blue")
hyper.drawRayFromVector(C, v, "Green")
hyper.drawPoints(A, B,C)
hyper.drawTangentVector(A,v,"red")
hyper.drawTangentVector(B,v,"blue")
hyper.drawTangentVector(C,v,"Green")
hyper.tikzEnd()
\end{luacode*}
```



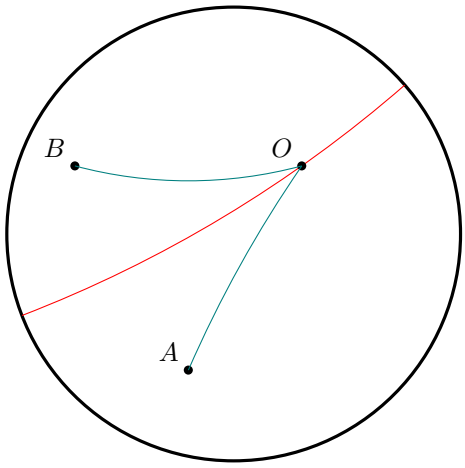
Perpendicular bisector

```
\begin{luacode*}
hyper.tikzBegin()
local A = complex(-0.2,-0.5)
local B = complex(-0.7,0.3)
hyper.drawPerpendicularBisector(A,B,"red")
hyper.drawPoints(A,B)
hyper.drawSegment(A, B, "teal")
hyper.labelPoints(A,"$A$", B, "$B$", "above
    ↪ right")
hyper.tikzEnd()
\end{luacode*}
```



Angle bisector

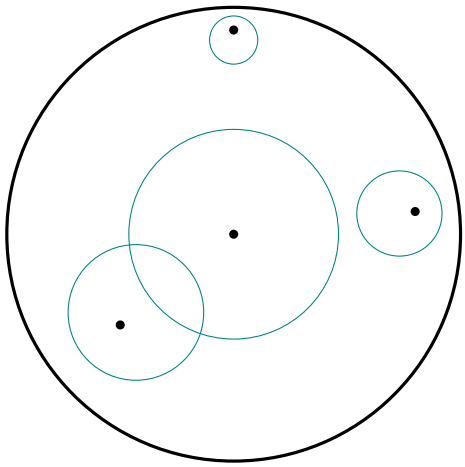
```
\begin{luacode*}
hyper.tikzBegin()
local A = complex(-0.2,-0.6)
local O = complex(0.3,0.3)
local B = complex(-0.7,0.3)
hyper.drawAngleBisector(A,O,B,"red")
hyper.drawPoints(A,O,B)
hyper.drawPolyline(A,O,B, "teal")
hyper.labelPoints(A, "$A$", O, "$O$", B, "$B$")
hyper.tikzEnd()
\end{luacode*}
```



2.3 Circles, semicircles, arcs

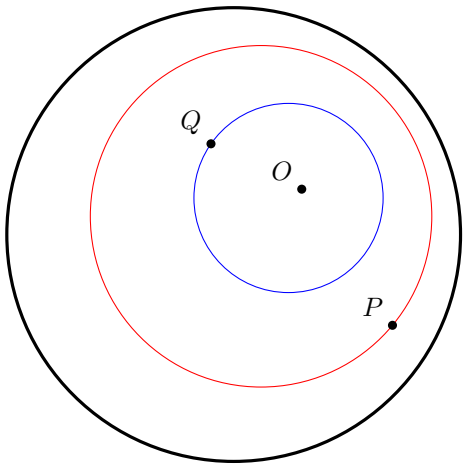
Center and radius

```
\begin{luacode*}
hyper.tikzBegin()
local points = {
  complex(0,0),
  complex(-0.5,-0.4),
  complex(0.8,0.1),
  complex(0,0.9)
}
for _,point in ipairs(points) do
  hyper.drawCircle(point,1,"teal")
  hyper.drawPoint(point)
end
-- NOTE : all circles have same radius=1 !
hyper.tikzEnd()
\end{luacode*}
```



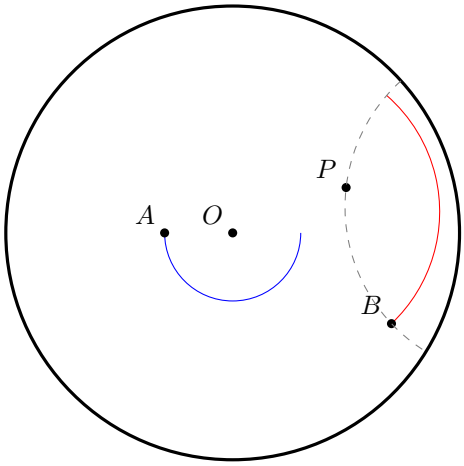
Circle through point

```
\begin{luacode*}
hyper.tikzBegin()
local O = complex(0.3,0.2)
local P = complex(0.7,-0.4)
local Q = complex(-0.1,0.4)
hyper.drawCircleThrough(O, P, "red")
-- or by hand :
hyper.drawCircle(O, hyper.distance(O,Q), "blue"
    ↪ )
hyper.drawPoints(O, P, Q)
hyper.labelPoints(O, "$O$", P, "$P$", Q, "$Q$")
hyper.tikzEnd()
\end{luacode*}
```



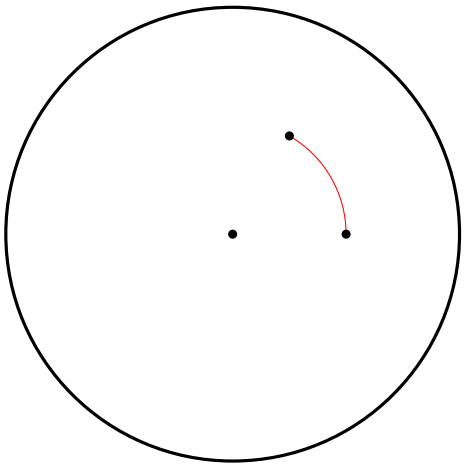
Semicircles

```
\begin{luacode*}
hyper.tikzBegin()
local O = complex(0,0)
local A = complex(-0.3,0)
local P = complex(0.5,0.2)
local B = complex(0.7,-0.4)
hyper.drawSemicircle(O, A, "blue")
hyper.drawSemicircle(P, B, "red")
hyper.drawPoints(O, A, P, B)
hyper.labelPoints(O, "$O$", A, "$A$", P, "$P$",
    ↪  B, "$B$")
hyper.drawLine(P,B, "gray, dashed")
hyper.tikzEnd()
\end{luacode*}
```



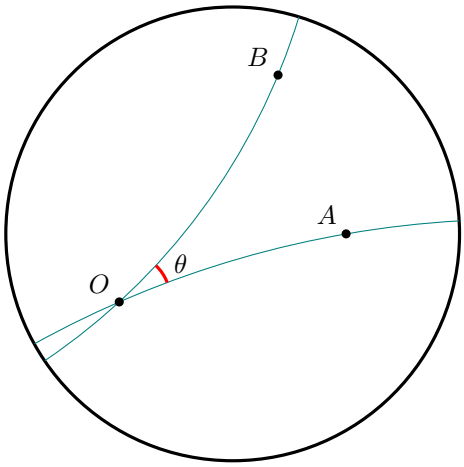
Arc (center and endpoints)

```
\begin{luacode*}
hyper.tikzBegin()
local O = complex(0,0)
local A = complex(0.5,0)
local B = complex.polar(0.5,math.pi/3)
hyper.drawArc(O,A,B, "red")
hyper.drawPoints(O,A,B)
hyper.tikzEnd()
\end{luacode*}
```



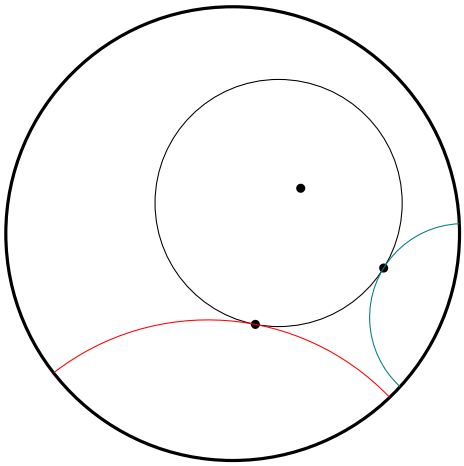
Draw angle $\widehat{AOB}$

```
\begin{luacode*}
hyper.tikzBegin()
local O = complex(-.5,-.3)
local A = complex(0.5,0)
local B = complex(0.2,.7)
hyper.drawLines(O, A, O, B, "teal")
hyper.drawAngle(A,O,B, "red, very thick")
hyper.drawPoints(A,O,B)
hyper.labelPoints(A, "$A$", O, "$O$", B, "$B$")
-- poor man's labelAngle:
hyper.labelPoint(O,"$\\theta$", "above=.5cm,
    ↪ right=.6cm")
hyper.tikzEnd()
\end{luacode*}
```



Tangent at point

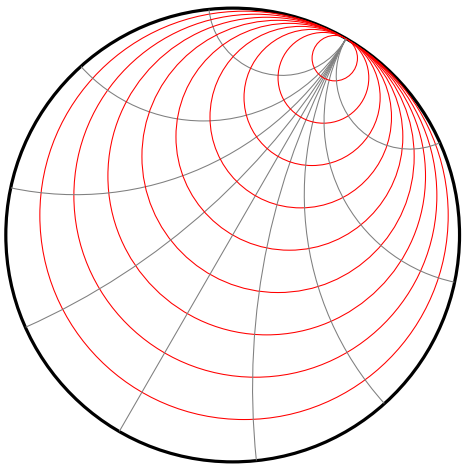
```
\begin{luacode*}
hyper.tikzBegin()
local O = complex(0.3,0.2)
local P = complex(0.1,-0.4)
local Q = hyper.rotation(O,math.pi/3)(P)
hyper.drawCircleThrough(O, P)
hyper.drawPoints(O, P, Q)
hyper.drawTangentAt(O,P,"red")
hyper.drawTangentAt(O,Q,"teal")
hyper.tikzEnd()
\end{luacode*}
```



2.4 Horocycles

Horocycles

```
\begin{luacode*}
hyper.tikzBegin()
local theta=math.pi/3
local idealPoint = complex.exp_i(theta)
local N = 10
for k=1,N-1 do
  local point = idealPoint * (2*k/N-1)
  hyper.drawHorocycle(idealPoint,point,"red")
  -- and an orthogonal geodesic :
  local endPoint = complex.exp_i(theta+2*k*math
      ↪ .pi/N)
  hyper.drawLine(idealPoint,endPoint,"gray")
end
hyper.tikzEnd()
\end{luacode*}
```

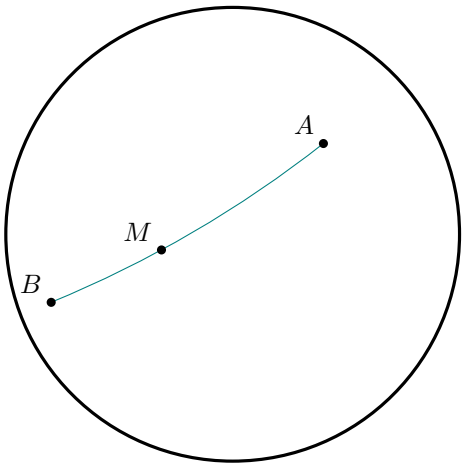


3 Defining points

3.1 Midpoint, interpolate, barycenter

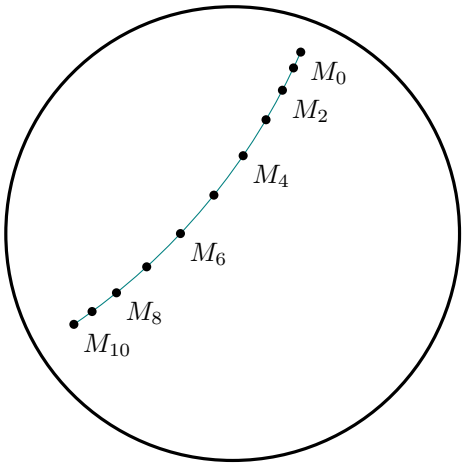
Midpoint

```
\begin{luacode*}
hyper.tikzBegin()
local A = complex(0.4,0.4)
local B = complex(-0.8,-0.3)
local M = hyper.midpoint(A,B)
hyper.drawSegment(A,B, "teal")
hyper.drawPoints(A,B,M)
hyper.labelPoints(A, "$A$", B, "$B$", M, "$M$")
hyper.tikzEnd()
\end{luacode*}
```



Interpolate and barycenter

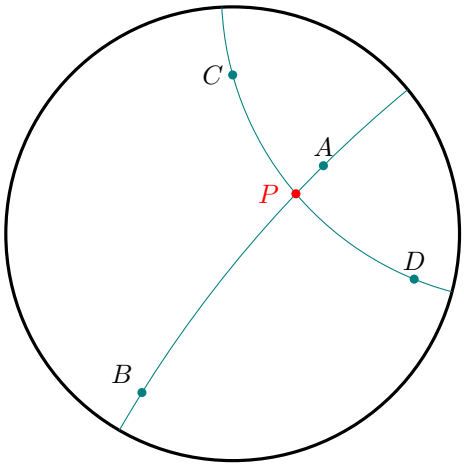
```
\begin{luacode*}
hyper.tikzBegin()
local A = complex(0.3,0.8)
local B = complex(-0.7,-0.4)
hyper.drawSegment(A,B, "teal")
for k=0,10 do
  local M = hyper.interpolate(A, B, k/10)
  hyper.drawPoint(M)
  if k % 2 == 0 then
    hyper.labelPoint(M, "$M_{" .. k .. "}$", "
        ↪ below right")
  end
end
hyper.tikzEnd()
\end{luacode*}
```



3.2 Intersections

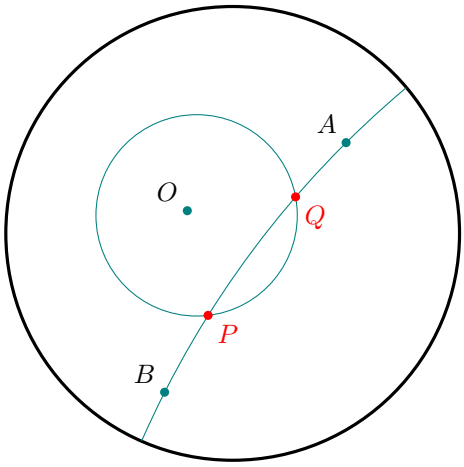
Line-Line (LL) intersection

```
\begin{luacode*}
hyper.tikzBegin()
local A = complex(0.4,0.3)
local B = complex(-0.4,-0.7)
local C = complex(0,0.7)
local D = complex(0.8,-0.2)
local P = hyper.interLL(A, B, C, D)
hyper.drawLines(A, B, C, D, "teal")
hyper.drawPoints(A, B, C, D,"teal")
hyper.drawPoint(P,"red")
hyper.labelPoint(A, "$A$", "above")
hyper.labelPoint(B, "$B$")
hyper.labelPoint(C, "$C$", "left")
hyper.labelPoint(D, "$D$", "above")
hyper.labelPoint(P, "$P$", "left=.1cm, red")
hyper.tikzEnd()
\end{luacode*}
```



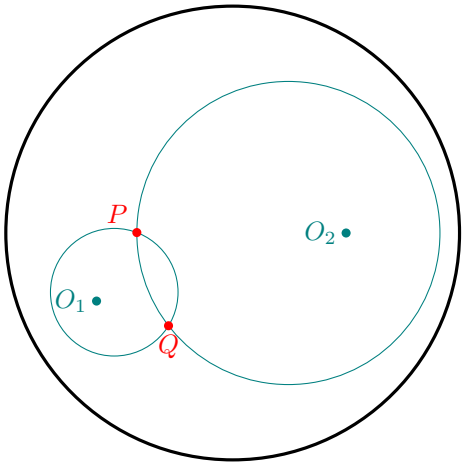
Line-circle (LC) intersection

```
\begin{luacode*}
hyper.tikzBegin()
local A = complex(0.5,0.4)
local B = complex(-0.3,-0.7)
local O = complex(-0.2,0.1)
local radius = 1
local P, Q = hyper.interLC(A, B, O, radius)
hyper.drawLine(A, B, "teal")
hyper.drawCircle(O,radius, "teal")
hyper.drawPoints(A, B, O, "teal")
hyper.drawPoints(P, Q, "red")
hyper.labelPoints(A, "$A$", B, "$B$", O, "$O$")
hyper.labelPoints(P, "$P$", Q, "$Q$", "red,
    ↪ below right")
hyper.tikzEnd()
\end{luacode*}
```



Circle-circle (CC) intersection

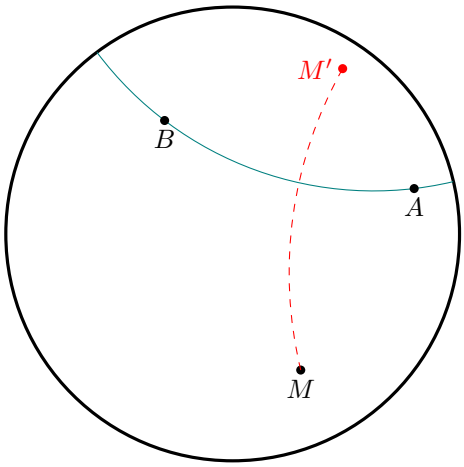
```
\begin{luacode*}
hyper.tikzBegin()
local O1 = complex(-0.6,-.3)
local O2 = complex(0.5,0)
local r1, r2 = 1, 2
local P, Q = hyper.interCC(O1, r1, O2, r2)
hyper.drawCircle(O1,r1, "teal")
hyper.drawCircle(O2,r2, "teal")
hyper.drawPoints(P, Q, "red")
hyper.drawPoints(O1, O2, "teal")
hyper.labelPoints(O1, "$O_1$", O2, "$O_2$", "
    ↪ left, teal")
hyper.labelPoints(P, "$P$", "above left, red")
hyper.labelPoint(Q, "$Q$", "red, below")
hyper.tikzEnd()
\end{luacode*}
```



3.3 Reflections and projections

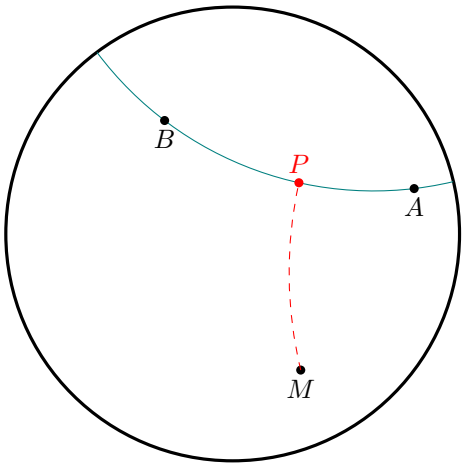
Reflection across a geodesic line

```
\begin{luacode*}
hyper.tikzBegin()
local A = complex(0.8,0.2)
local B = complex(-0.3,0.5)
local M = complex(0.3,-0.6)
local Mp = (hyper.reflection(A,B))(M)
hyper.drawLine(A, B, "teal")
hyper.drawPoints(A, B, M)
hyper.drawPoint(Mp,"red")
hyper.drawSegment(M,Mp,"dashed,red")
hyper.labelPoint(Mp,"$M'$", "left, red")
hyper.labelPoints(A, "$A$", B, "$B$", M, "$M$",
    ↪  "below")
hyper.tikzEnd()
\end{luacode*}
```



Projection onto a geodesic line

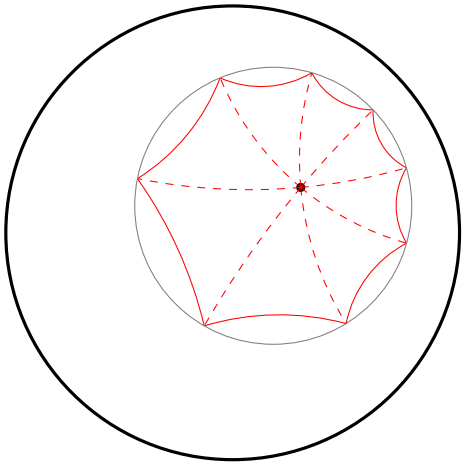
```
\begin{luacode*}
hyper.tikzBegin()
local A = complex(0.8,0.2)
local B = complex(-0.3,0.5)
local M = complex(0.3,-0.6)
local P = (hyper.projection(A,B))(M)
hyper.drawLine(A, B, "teal")
hyper.drawPoints(A, B, M)
hyper.drawPoint(P,"red")
hyper.drawSegment(M,P,"dashed,red")
hyper.labelPoint(P,"$P$", "above, red")
hyper.labelPoints(A, "$A$", B, "$B$", M, "$M$",
    ↪  "below")
hyper.tikzEnd()
\end{luacode*}
```



3.4 (Elliptic) Rotations

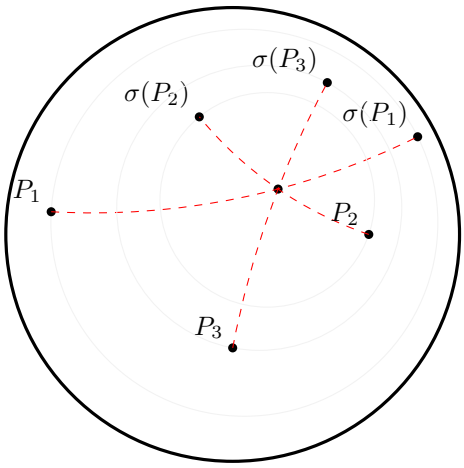
(Elliptic) rotation around a point (regular octogon)

```
\begin{luacode*}
hyper.tikzBegin()
local O = complex(0.3,0.2)
local M = complex(0.5,-0.4)
hyper.drawPoints(O)
for k=1,8 do
  local P = (hyper.rotation(O,2*math.pi/8))(M)
  hyper.drawSegment(M, P, "red")
  hyper.drawSegment(O,M, "dashed,red")
  M = P
end
hyper.drawCircle(O,hyper.distance(O,M),"thin,
    ↪ gray")
hyper.tikzEnd()
\end{luacode*}
```



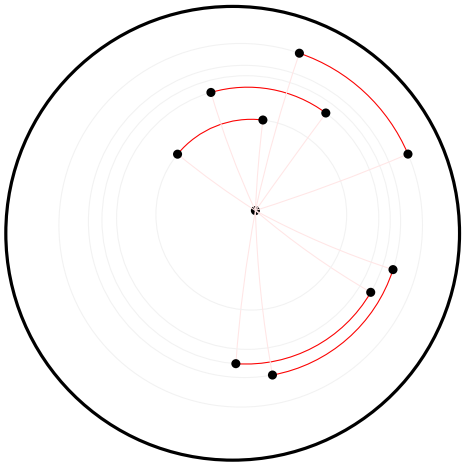
Symmetry around a point (rotation by 180°)

```
\begin{luacode*}
hyper.tikzBegin()
local O = complex(0.2,0.2)
local s = hyper.symmetry(O)
hyper.drawPoints(O)
local points={complex(-.8,.1), complex(.6,0),
    ↪ complex(0,-.5)}
for k, P in ipairs(points) do
  hyper.drawCircleThrough(O, P, "gray!10")
  local sP = s(P)
  hyper.drawPoints(P,sP)
  hyper.drawSegment(P,sP, "red, dashed")
  hyper.labelPoint(P, "$P_{".. k .."}$")
  hyper.labelPoint(sP, "$\\sigma(P_{".. k .."})
      ↪ $")
end
hyper.tikzEnd()
\end{luacode*}
```



Rotation by 60°

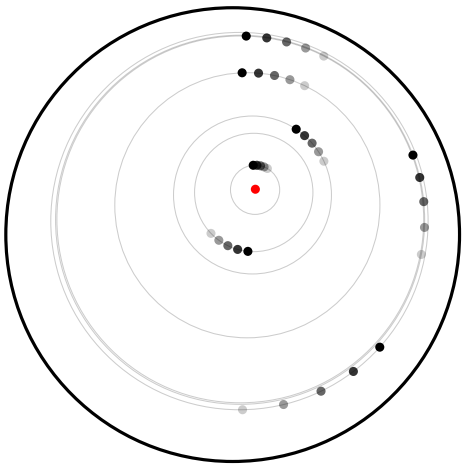
```
\begin{luacode*}
hyper.tikzBegin()
local O = complex(0.1,0.1)
local r = hyper.rotation(O,math.pi/3)
hyper.drawPoints(O)
for k=1,5 do
  local P = hyper.randomPoint(0.3,0.9)
  local rP = r(P)
  hyper.drawCircleThrough(O,P, "gray!10")
  hyper.drawArc(O,P,rP,"red")
  hyper.drawPolyline(P,O,rP, "red!10")
  hyper.drawPoints(P,rP)
end
hyper.tikzEnd()
\end{luacode*}
```



3.5 Orbits under automorphisms

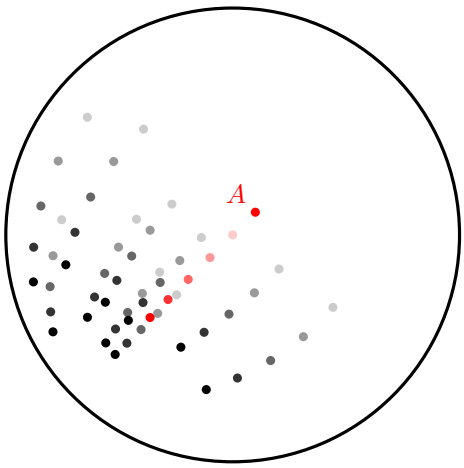
Orbit under elliptic automorphism (rotation)

```
\begin{luacode*}
hyper.tikzBegin()
local nbPoints = 7
local nbIterations = 5
local O = complex(0.1,0.2)
local rot = hyper.rotation(O,math.pi/20) --
    ↪ angle in radians
hyper.drawPoint(O,"red")
for k=1,nbPoints do
  local P = hyper.randomPoint(0,.9)
  hyper.drawCircleThrough(O,P,"draw opacity=0.2
      ↪ ")
  hyper.drawPointOrbit(P, rot, nbIterations)
end
hyper.tikzEnd()
\end{luacode*}
```



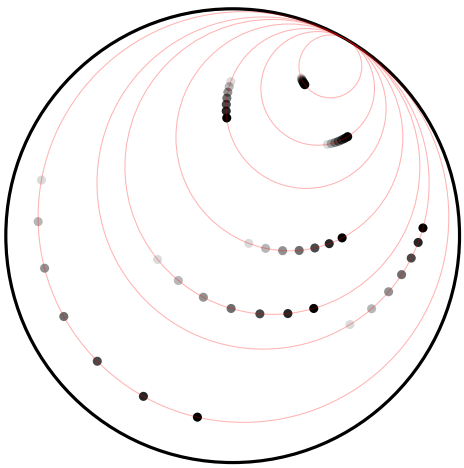
Orbit under hyperbolic automorphism

```
\begin{luacode*}
hyper.tikzBegin()
local nbPoints = 10
local nbIterations = 5
local A = complex(0.1,0.1)
local phi = hyper.automorphism(A,0)
for k=1,nbPoints do
  local P = hyper.randomPoint(0,.9)
  hyper.drawPointOrbit(P, phi, nbIterations)
end
hyper.drawPoint(A,"red")
hyper.labelPoint(A,"$A$", "above left, red")
hyper.drawPointOrbit(A, phi, nbIterations, "red
    ↪ ")
hyper.tikzEnd()
\end{luacode*}
```



Orbit under parabolic automorphism

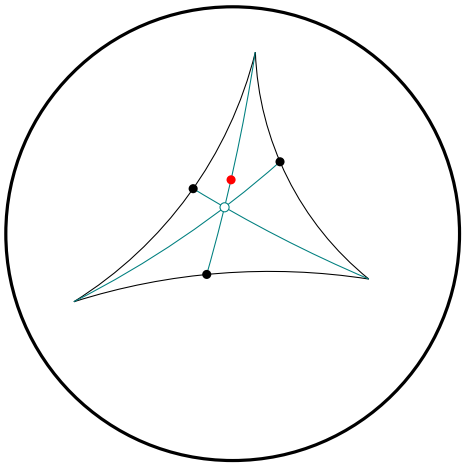
```
\begin{luacode*}
hyper.tikzBegin()
local nbIterations = 7
limit = complex.exp_i(math.pi/3) -- on circle
local phi = hyper.parabolic(limit,math.pi/20)
local points={
  complex(0,0), complex(0,.7),
  complex(.4,.4), complex(.3,.7),
  complex(-.8,.4), complex(-.4,0),
  complex(.4,-.45)}
for _,P in ipairs(points) do
  hyper.drawPointOrbit(P, phi, nbIterations)
  hyper.drawHorocycle(limit, P, "red, draw
      ↪ opacity=.3")
end
hyper.tikzEnd()
\end{luacode*}
```



3.6 Triangle geometry (CAUTION)

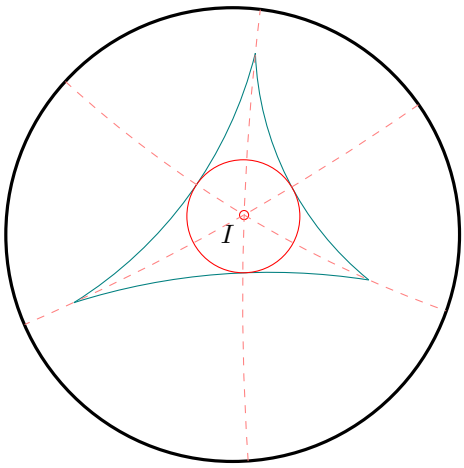
Centroid (intersection of medians)

```
\begin{luacode*}
hyper.tikzBegin()
local A = complex(0.1,0.8)
local B = complex(-0.7,-0.3)
local C = complex(0.6,-0.2)
local AA = hyper.midpoint(B,C)
local BB = hyper.midpoint(C,A)
local CC = hyper.midpoint(A,B)
local G = hyper.triangleCentroid(A,B,C)
hyper.drawTriangle(A,B,C)
hyper.drawSegments(A, AA, B, BB, C, CC, "teal")
hyper.drawPoints(AA,BB,CC)
hyper.drawPoint(G,"white, draw=teal")
-- CAUTION :
local wrong = hyper.interpolate(A,AA,2/3)
hyper.drawPoint(wrong, "red")
hyper.tikzEnd()
\end{luacode*}
```



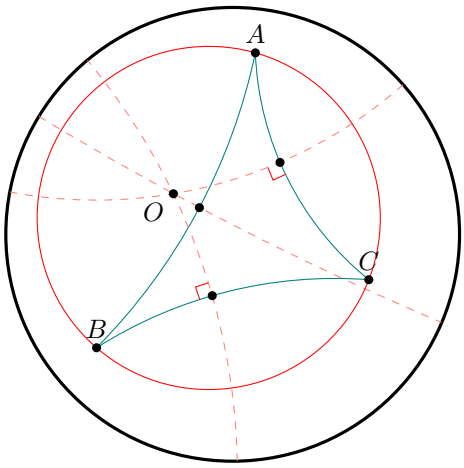
Incenter and incircle

```
\begin{luacode*}
hyper.tikzBegin()
local A = complex(0.1,0.8)
local B = complex(-0.7,-0.3)
local C = complex(0.6,-0.2)
local I = hyper.triangleIncenter(A,B,C)
hyper.drawTriangle(A,B,C,"teal")
hyper.drawPoint(I,"white, draw=red")
hyper.labelPoint(I,"$I$", "below left")
hyper.drawIncircle(A,B,C,"red")
-- show intersection :
local mystyle = "dashed, red!50"
hyper.drawAngleBisector(A,B, C, mystyle)
hyper.drawAngleBisector(B,C,A, mystyle)
hyper.drawAngleBisector(C,A,B, mystyle)
hyper.tikzEnd()
\end{luacode*}
```



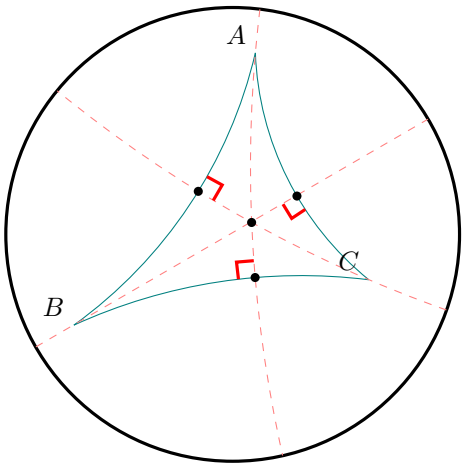
Circumcenter and circumcircle (CAUTION : does not always exist)

```
\begin{luacode*}
hyper.tikzBegin()
local A = complex(0.1,0.8)
local B = complex(-0.6,-0.5)
local C = complex(0.6,-0.2)
hyper.drawTriangle(A,B,C,"teal")
local mystyle = "dashed, red!50"
hyper.drawPerpendicularBisector(A, B, mystyle)
hyper.drawPerpendicularBisector(B, C, mystyle)
hyper.drawPerpendicularBisector(C, A, mystyle)
local O = hyper.triangleCircumcenter(A,B,C)
hyper.drawCircumcircle(A,B,C,"red")
hyper.labelPoint(O,"$O$", "below left")
local BC = hyper.midpoint(B,C)
local CA = hyper.midpoint(C,A)
local AB = hyper.midpoint(A,B)
hyper.drawRightAngle(O, CA, C,"red",1/8)
hyper.drawRightAngle(O,BC,B,"red",1/8)
hyper.drawPoints(O,A,B,C,AB,BC,CA)
hyper.labelPoints(A, "$A$", B, "$B$", C, "$C$",
    ↪ "above")
hyper.tikzEnd()
\end{luacode*}
```



Orthocenter (CAUTION : does not always exist)

```
\begin{luacode*}
hyper.tikzBegin()
local A = complex(0.1,0.8)
local B = complex(-0.7,-0.4)
local C = complex(0.6,-0.2)
hyper.drawTriangle(A,B,C,"teal")
local mystyle = "dashed, red!50"
hyper.drawPerpendicularThrough(A,B,C,mystyle)
hyper.drawPerpendicularThrough(B,C,A,mystyle)
hyper.drawPerpendicularThrough(C,A,B,mystyle)
local AA = hyper.projection(B,C)(A)
local BB = hyper.projection(C,A)(B)
local CC = hyper.projection(A,B)(C)
local style2 = "very thick, red"
hyper.drawRightAngle(A,AA,B,style2,1/15)
hyper.drawRightAngle(B,BB,C,style2,1/8)
hyper.drawRightAngle(C,CC,A,style2,1/12)
hyper.labelPoints(A, "$A$", B, "$B$", C, "$C$")
local H = hyper.triangleOrthocenter(A,B,C)
hyper.drawPoints(AA,BB,CC,H,"above")
hyper.tikzEnd()
\end{luacode*}
```

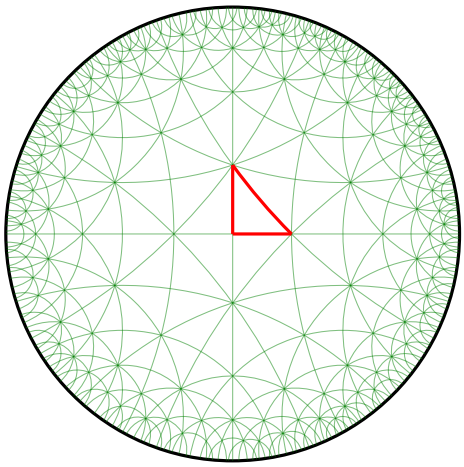


4 Tilings

The two following tilings are drawn with `TILING_DEPTH=12`

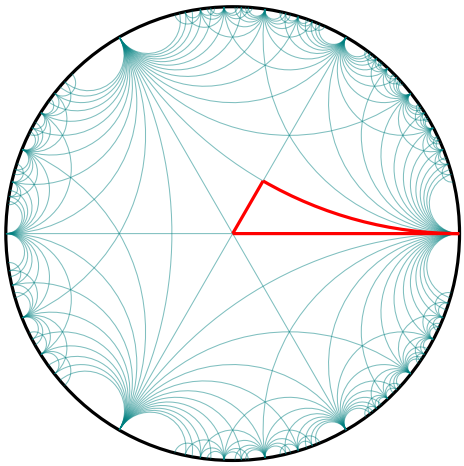
Triangle tiling

```
\begin{luacode*}
hyper.tikzBegin()
local A, B, C = hyper.fundamentalRightTriangle
    ↪ (4,5)
local geodesics = {
{hyper.endpoints(A,B)},
  {hyper.endpoints(B,C)},
  {hyper.endpoints(C,A)}}
geodesics = hyper.propagate_geodesics(geodesics
    ↪ , TILING_DEPTH)
hyper.drawLinesFromTable(geodesics, "green!50!
    ↪ black, draw opacity=.5")
hyper.drawTriangle(A,B,C,"red, very thick")
hyper.tikzEnd()
\end{luacode*}
```



Triangle tiling with one angle zero

```
\begin{luacode*}
hyper.tikzBegin()
local A, B, C = hyper.fundamentalIdealTriangle
    ↪ (3,2)
local geodesics = {
  {hyper.endpoints(A,B)},
  {hyper.endpoints(B,C)},
  {hyper.endpoints(C,A)}
}
geodesics = hyper.propagate_geodesics(geodesics
    ↪ , TILING_DEPTH)
hyper.drawLinesFromTable(geodesics, "teal, draw
    ↪  opacity=.5")
hyper.drawTriangle(A,B,C,"red, very thick")
hyper.tikzEnd()
\end{luacode*}
```

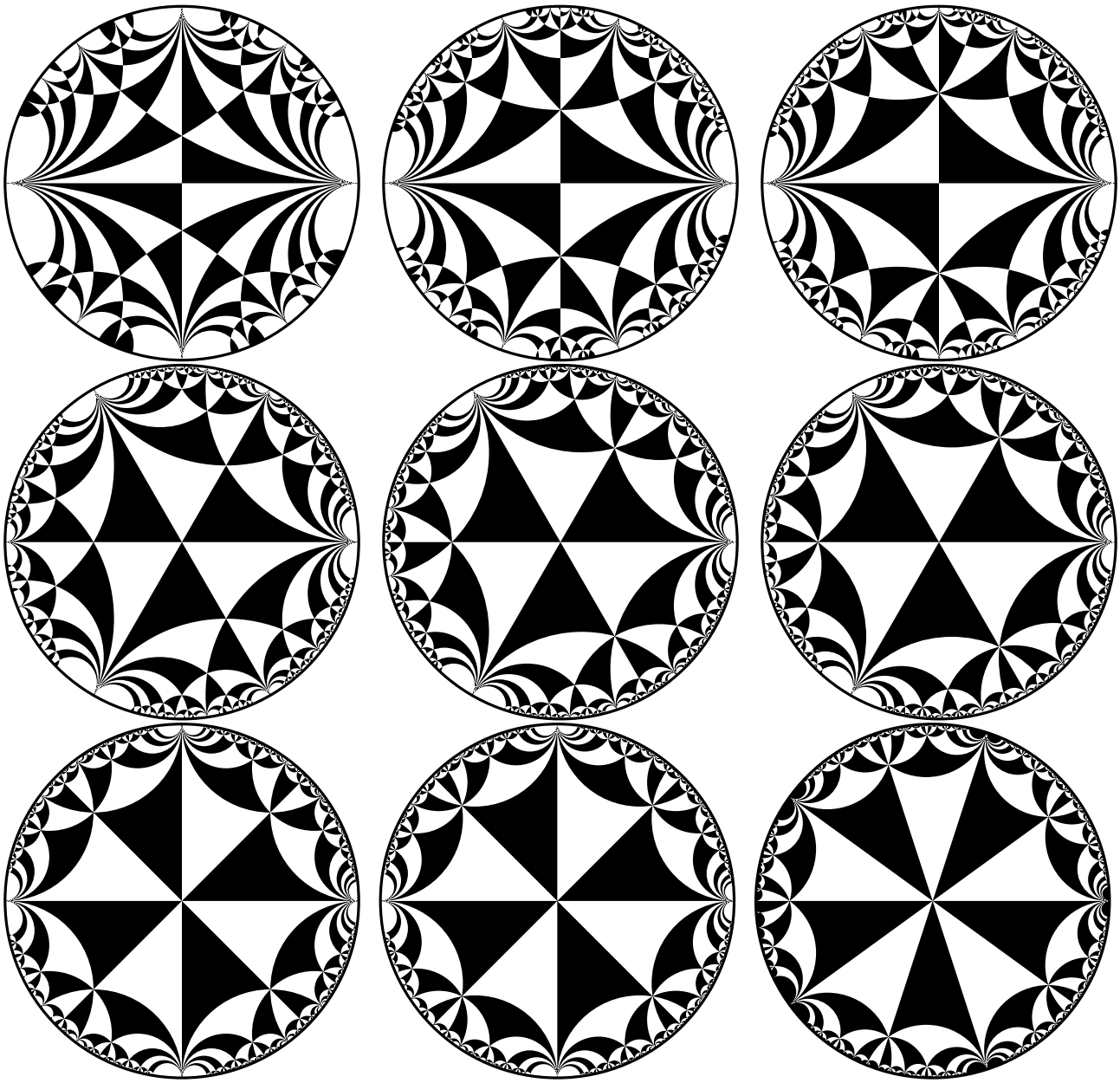


4.1 Source code of first picture

```
\begin{luacode*}
local A, B, C = hyper.fundamentalRightTriangle(4,5)
local phi = hyper.automorphism(C,0)
A,B,C = phi(A), phi(B), phi(C) -- center on pentagon
local geodesics = {
  {hyper.endpoints(A,B)},
  {hyper.endpoints(B,C)},
  {hyper.endpoints(C,A)}
}
geodesics = hyper.propagate_geodesics(geodesics, 24)
local shapes={}
for k,pair in ipairs(geodesics) do
  table.insert(shapes,hyper.shape_closed_line(pair[1], pair[2]))
end
shapes_string = table.concat(shapes, " ")
hyper.tikzBegin("scale=6")
hyper.tikzPrintf("\\fill[even odd rule] (0,0) circle (1) %s ;",shapes_string)
hyper.drawRegularPolygon(complex.ZERO, B,5,  "line width=.15cm, cyan")
```

```
hyper.tikzEnd()
\end{luacode*}
```

4.2 More images just for fun



5 Overview of the lua libraries

The Lua files are in the `lua/` subdirectory of the repo.

To build the `luahyperbolic.sty` from the lua source files, run `lua build.lua` on the root directory. (Lua must be installed)

The Lua source files are not needed to compile a latex document using luahyperbolic, only the sty file is necessary.

5.1 complex.lua

Functions

__add	__mul	__tostring	abs2	conj	dot
__div	__pow	__unm	arg	det	exp
__eq	__sub	abs	clone	distinct	exp_i

invert	isImag	isNot	new	rotate90
isClose	isInteger	isReal	nonzero	scal
isClosePair	isNear	isUnit	polar	toPolar
isColinear	isNearInteger	log	rotate	unit

Constants (numbers and complex numbers)

EPS, EPS_INV, I, J, ONE, ZERO, ___index,

5.2 luahyperbolic-tikz.lua and hyperbolic-core

Functions

acosh	drawPolygon	metric_factor
angle	drawPolygonFromTable	midpoint
asinh	drawPolyline	midpoint_to_origin
assert	drawPolylineFromTable	mobius_transformation
assert_in_disk	drawRay	on_circle
atanh	drawRayFromPoints	parabolic
automorphism	drawRayFromVector	parabolic_fix1
automorphism_from_pairs	drawRegularPolygon	projection
automorphism_sending	drawRightAngle	propagate_geodesics
barycenter2	drawSegment	radial_half
circle_to_euclidean	drawSegments	radial_scale
cosh	drawSemicircle	randomPoint
distance	drawTangentAt	reflection
distance_to_geodesic	drawTangentVector	rotation
distance_to_origin	drawTriangle	rotation_from_pair
drawAngle	endpoints	same_geodesics
drawAngleBisector	endpoints_angle_bisector	shape_closed_line
drawArc	endpoints_perpendicular_bisector	shape_euclidean_segment
drawCircle	ensure_order	shape_segment
drawCircleRadius	error	sinh
drawCircleThrough	exp_map	symmetry
drawCircumcircle	exp_map_at_origin	symmetry_around_midpoint
drawHorocycle	fundamentalIdealTriangle	tangentVector
drawIncircle	fundamentalRightTriangle	tikzBegin
drawLabeledPoint	geodesic_data	tikzClearBuffer
drawLine	in_closed_disk	tikzEnd
drawLineFromVector	in_closed_half_plane	tikzExport
drawLines	in_disk	tikzGetFirstLines
drawLinesFromTable	in_half_plane	tikzPrintf
drawPerpendicularBisector	interCC	tikz_define_nodes
drawPerpendicularThrough	interLC	triangleCentroid
drawPoint	interLL	triangleCircumcenter
drawPointOrbit	interpolate	triangleIncenter
drawPoints	is_on_geodesic	triangleOrthocenter
drawPolySeg	labelPoint	warning
drawPolySegFromTable	labelPoints	

Constants

Numbers : DRAW_EPS, DRAW_EPS_INV, DRAW_POINT_RADIUS, EPS, SCALE, tikzNbPicturesExported,

Strings : TIKZ_BEGIN_DISK, module, tikzOptions,

Tables : tikzBuffer,